

Project Title: Data-Driven Outreach Analysis and Campaign Optimization

Data Visualization Trainee Early Internship: Team-27



Presented by

Name	Roles	Email
Sumaiya Tasnim	Team Lead	sumaiyaa.tasnim.18@gmail.com
Wilgens Almonor	Project Scribe	Wilgensalmonor@gmail.com
Subashree J	Project Manager	subashreej03@gmail.com
Ayan Banik	Project Lead	ayanbanik001@gmail.com
Palak Garg	Team Member	palakgarg447@gmail.com
Asra Laraib Bhutta	Team Member	asralaraib1@gmail.com

Excelerate



PROJECT OBJECTIVES

OutreachData

Assigned
Datasets



CampaignData

ApplicantData

01

Dataset Cleaning

Handled Missing,
duplicate,inconsistent
values

05

Major Insights Identified

Key important insights were
recognized from Dashboard Analysis.

02

Joining Datasets

Ensuring accuracy, Joined the
datasets by identifying
primary & foreign keys

06

Recommendations

Analyzing the major insights, we
predicted recommendations for
optimizations

03

Exploratory Data Analysis

Analyzed & performed EDA of
the joined dataset to
understand the data insights

07

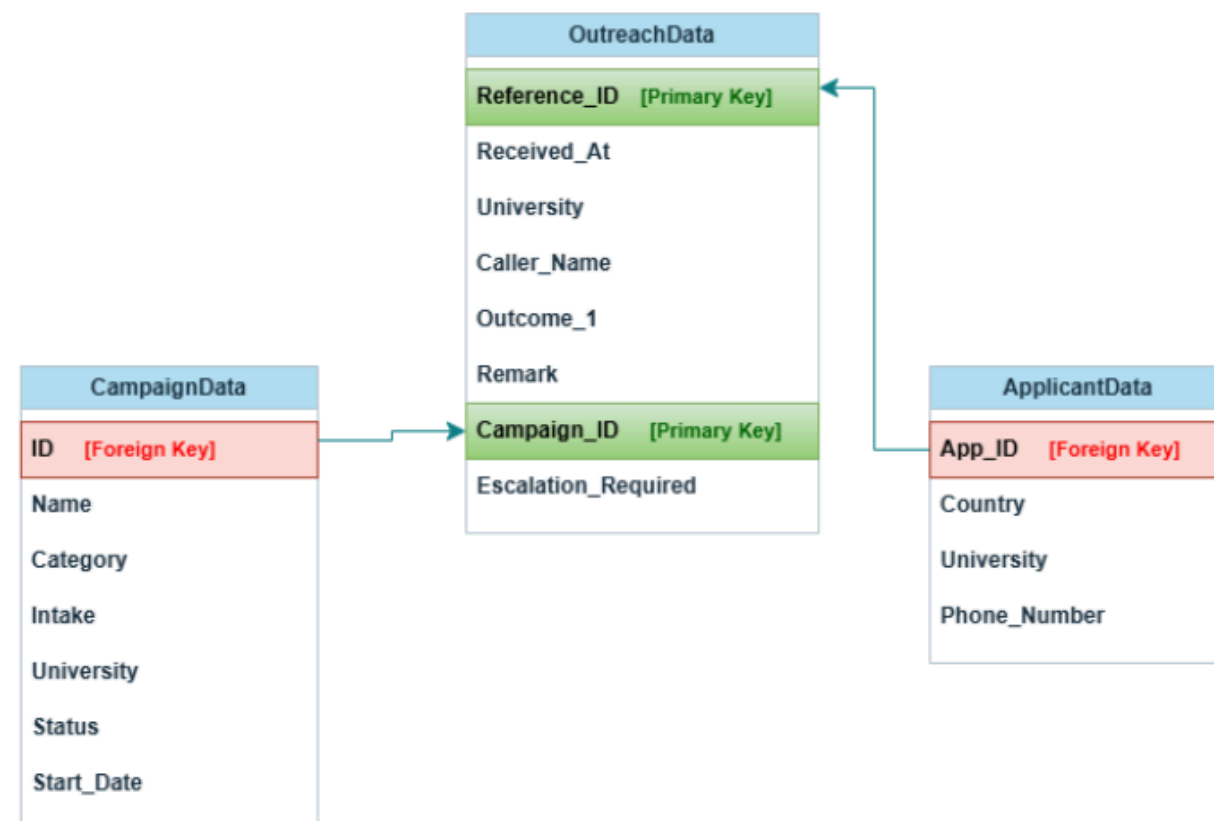
Action plan

Ready for Executing of
optimization plan

04

Interactive Dashboard Creation

Created an Interactive Dashboard including
important KPIs & Filters of Campaign
Outreach data Using POWER BI tool.



FINAL DASHBOARD



Dashboard Summary

KPIs: Tracks Total Outreach Attempts (40K), Response Rate (1.29%), Connectivity Rate (2.01 %), Avg Attempts per Response (77.42), and Escalation Rate (2.09%).

Filters: Global filters for Month, Country, Category, and Campaign ID.

Visuals: Includes Outreach vs Response Trends, Top Countries by Response Rate, Conversion by Escalation, Category-wise Response %, Weekly & Hourly Engagement Patterns, Outreach Volume Over Time, Phone Call Availability and Outcome Engagement Overview.

Insights: Provides a clear, data-driven view of outreach performance to support strategic decision-making and enrollment growth.

Outreach Campaign Performance Tracker Dashboard

Month
All

Country
All

Category
All

Campaign_ID
All

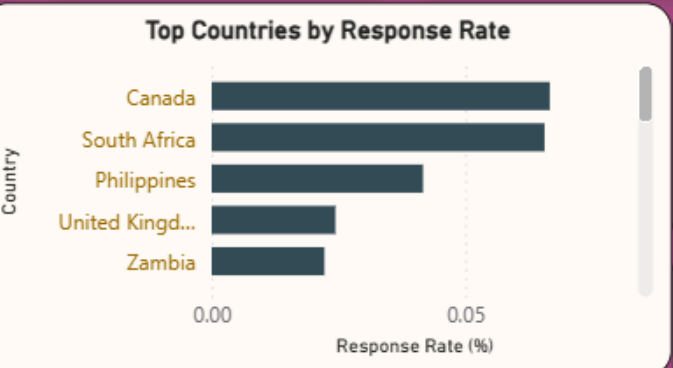
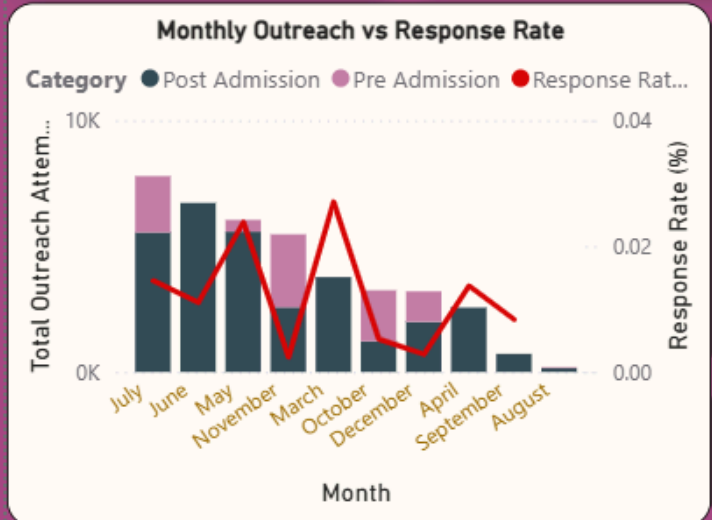
40K
Total Outreach Attempts

1.29%
Response Rate (%)

2.01%
Connectivity Rate (%)

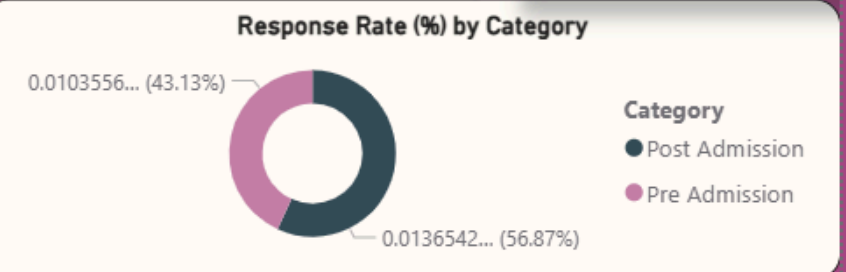
77.42
Avg Attempts per Response

2.09
Escalation Rate (%)



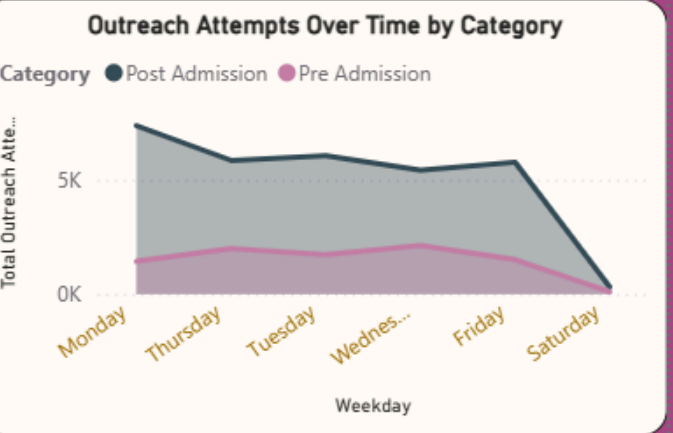
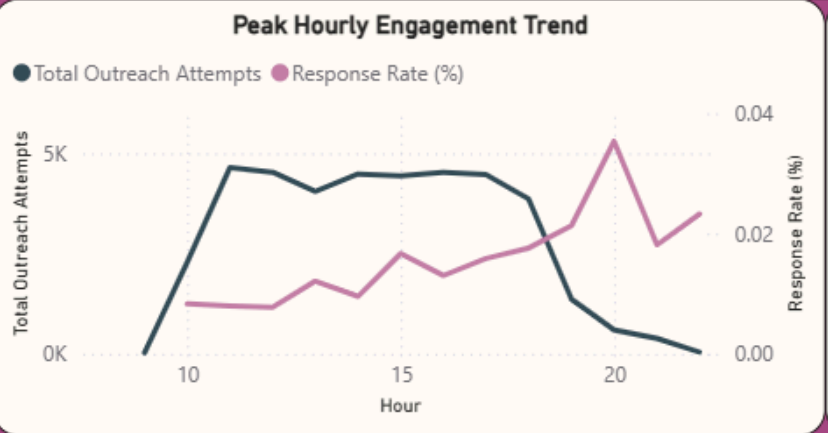
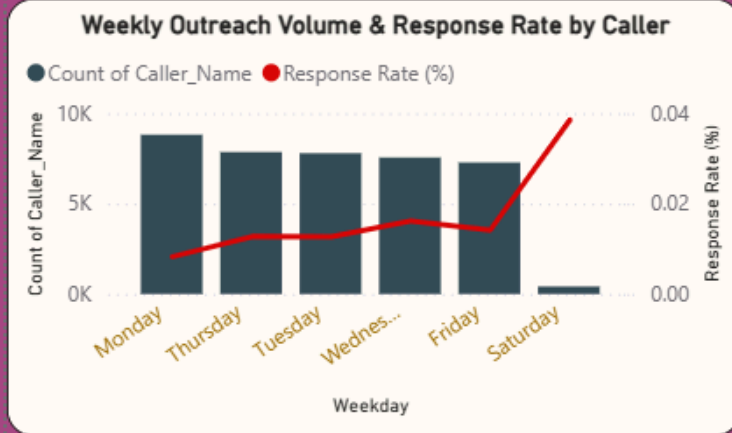
Outcome Engagement Status

8	27091
Connected	Not Connected
1543	1284
Disconnected	Wrong Number



Phone number Availability Overview

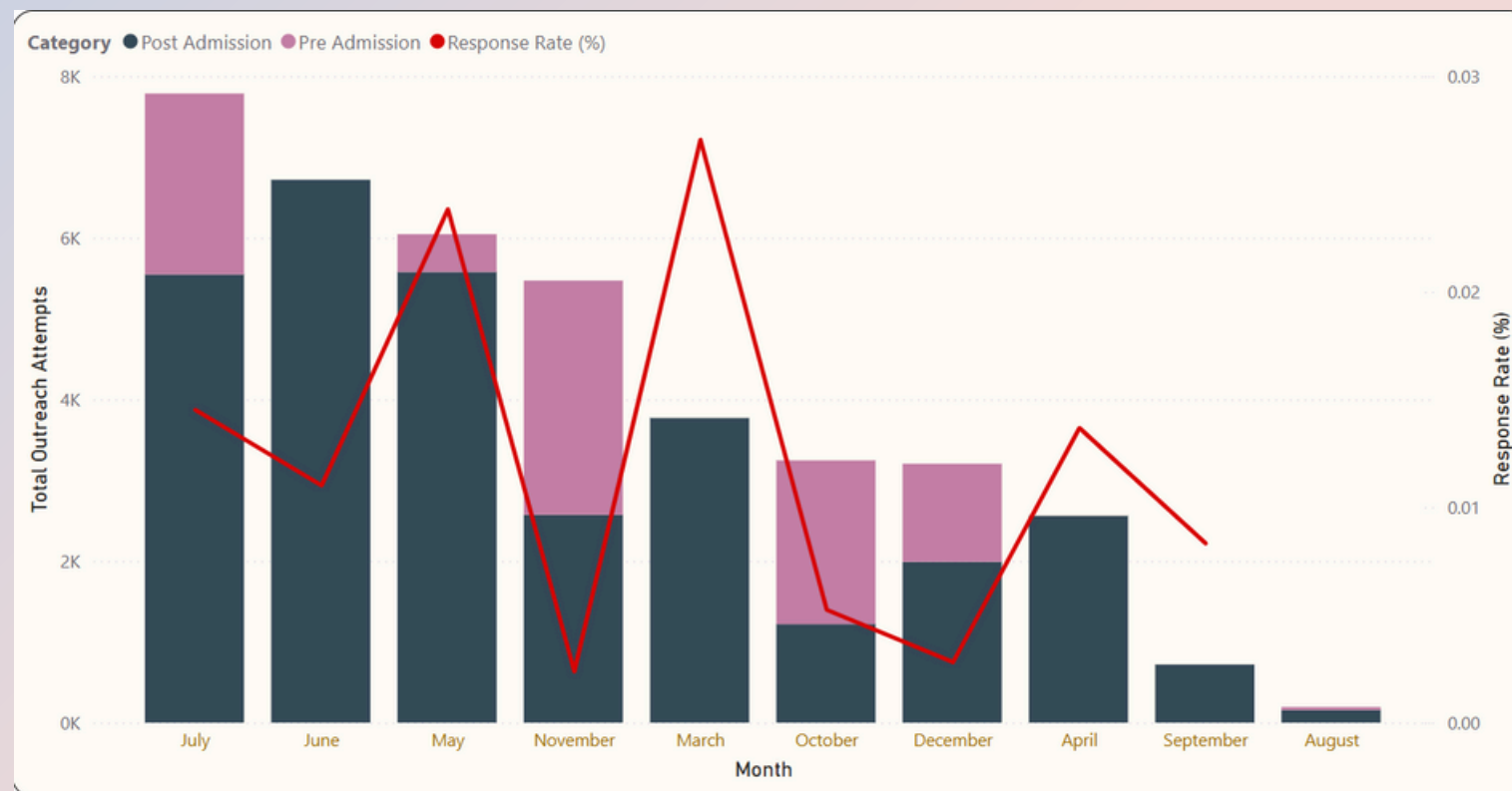
Sum of Available_numbers	Sum of Unavailable numbers
39458	259



KEY INSIGHTS



Monthly Outreach vs Response Rate



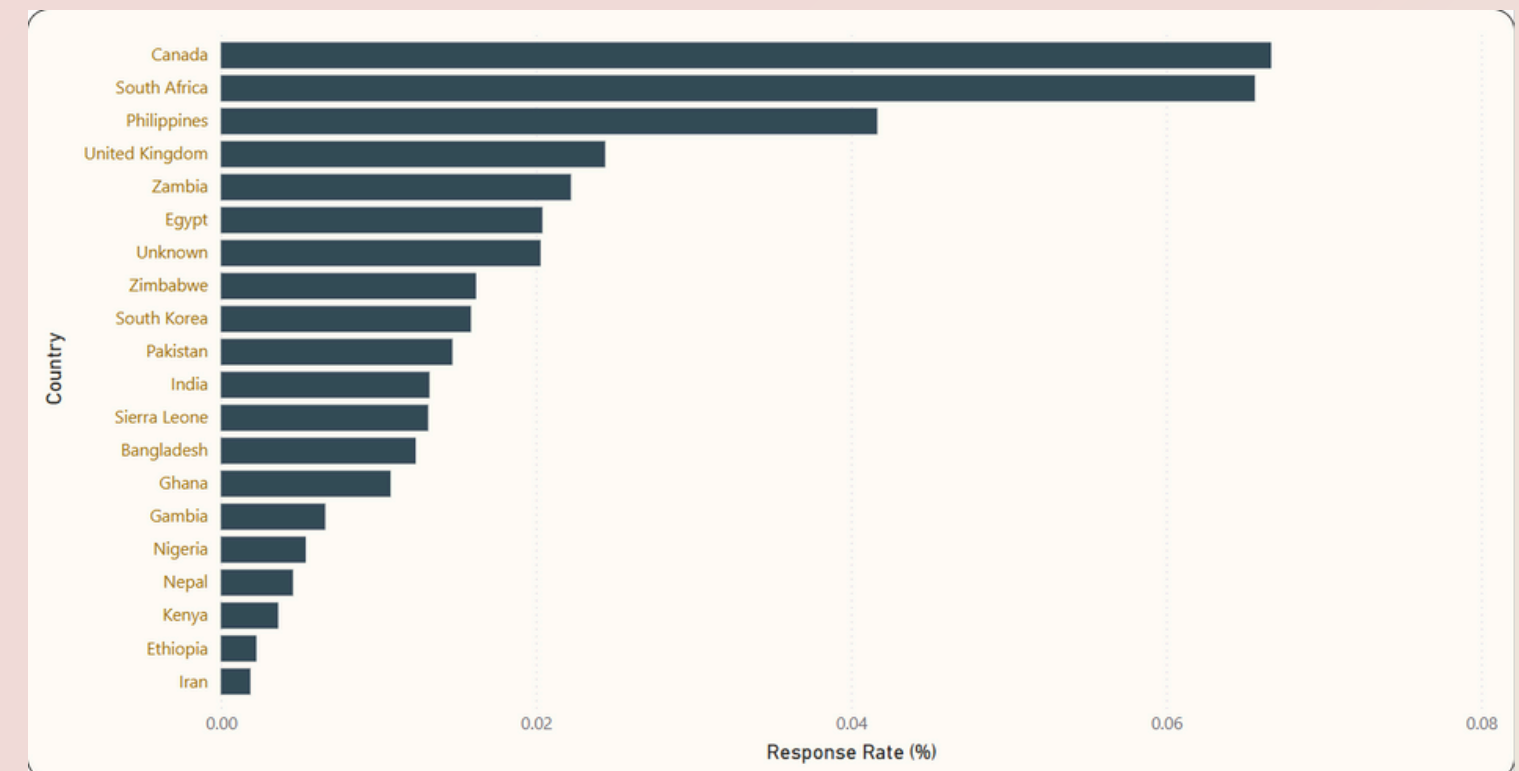
Insights:

The chart reveals that July had the highest total outreach attempts (both Pre and Post Admission), yet the highest Response Rate occurred in March, suggesting attempt volume isn't the sole driver of response success.

Recommendations

To maximize efficiency, prioritize outreach efforts in months with historically high response rates like March, and analyze the specific tactics used then for replication across all months.

Top Countries by Response Rate



Insights:

Canada and South Africa demonstrate the highest Response Rates, significantly outperforming all other countries, with Iran and Ethiopia showing the lowest engagement.

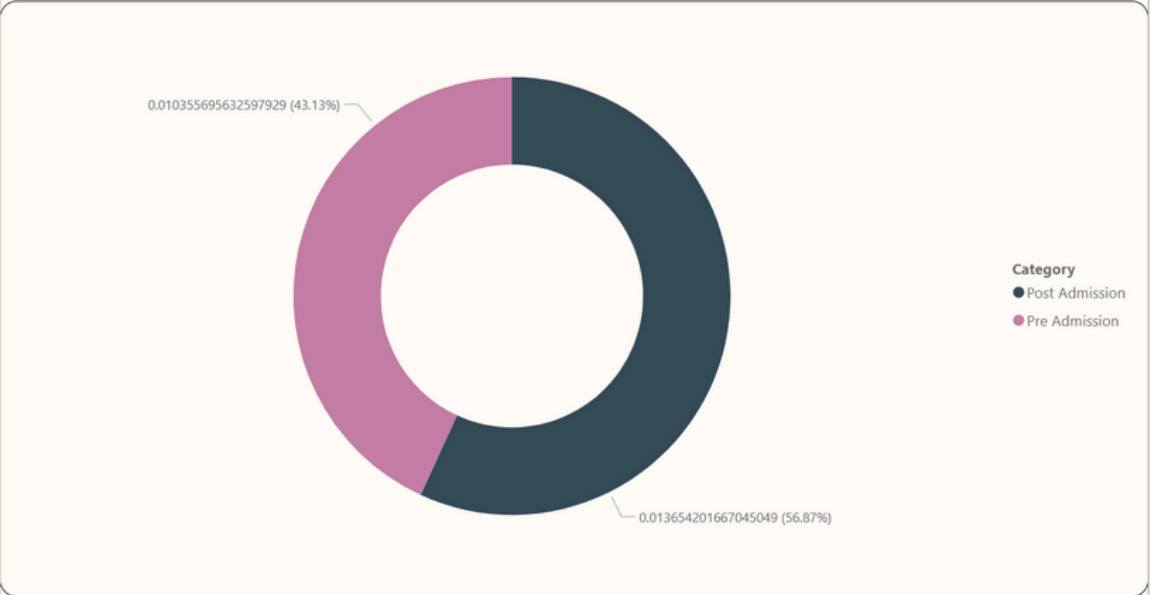
Recommendations

Focus increased outreach efforts and resource allocation on top-performing countries like Canada and South Africa, while analyzing the reasons for the extremely low response rates in Iran and Ethiopia to develop targeted engagement strategies.

KEY INSIGHTS



Response Rate by Category



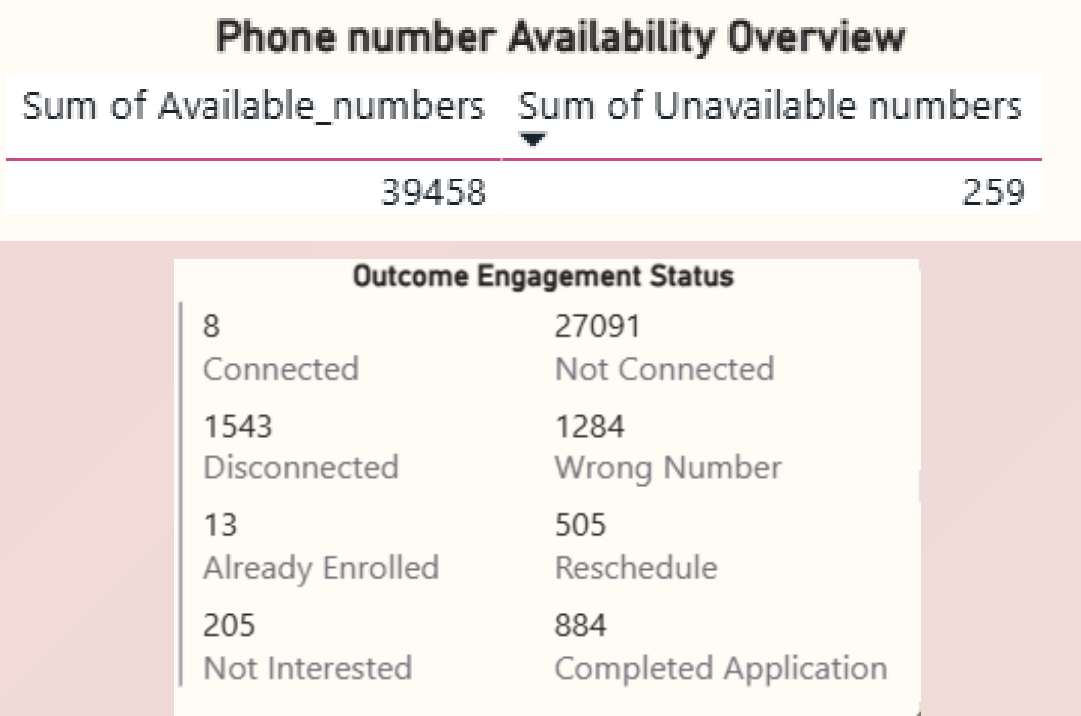
Insights:

The Post Admission category accounts for the majority of the total (approximately 56.87%), significantly exceeding the Pre Admission category which makes up about 43.13%.

Recommendations

To maximize overall success, focus analysis on the Post Admission category to understand the factors driving its greater magnitude, and identify opportunities to boost the smaller Pre Admission category's performance closer to parity.

Phone Call & Outcome Matrix Charts



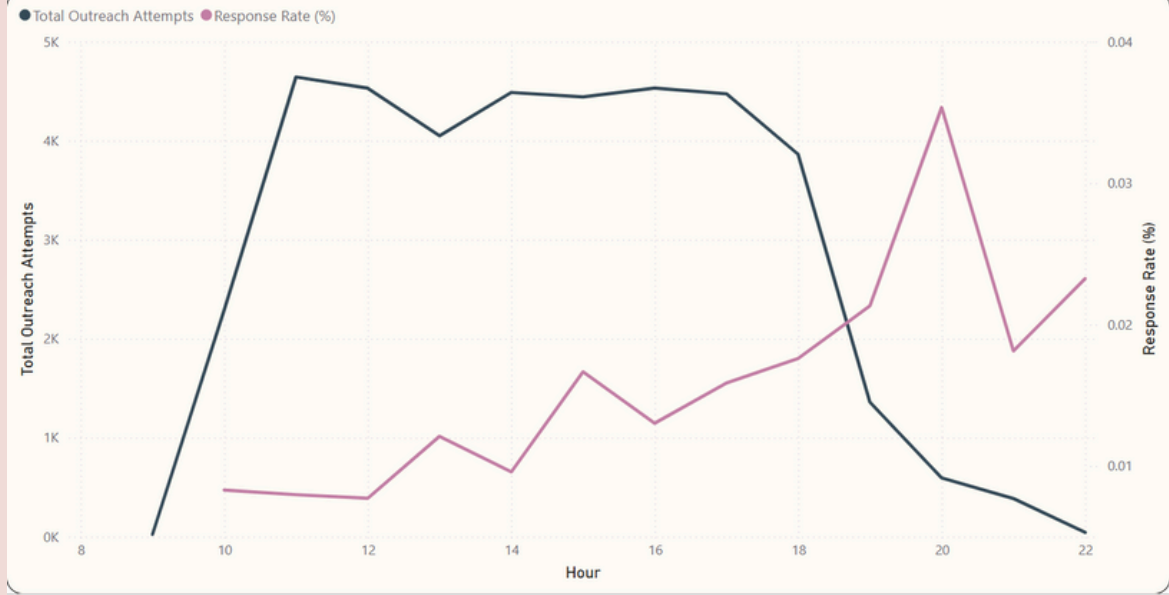
Insights:

- 📞 **Phone Call Connectivity:**
The vast majority of phone numbers are Available (39,458), with Unavailable numbers (259) representing a negligible fraction of the total.
- 🗣️ **Engagement Outcome:**
The vast majority of outreach attempts resulted in a "Not Connected" status (27,091), significantly overshadowing successful "Connected" outcomes (8) and other statuses.

Recommendation

- Continue to maintain high availability by monitoring for any increase in the small number of Unavailable phone numbers and proactively addressing any new issues immediately.
- Conduct an immediate investigation into the root causes of the extremely high "Not Connected" rate and implement strategies to improve contact rates, such as optimizing calling times or number quality.

Peak Hourly Engagement Trend



Insights:

The Total Outreach Attempts peak between 11 AM and 5 PM, while the Response Rate (%) follows a very different pattern, peaking significantly later around 8 PM, when attempts are low.

Recommendations

Shift the majority of outreach attempts to align with the peak Response Rate hours (7 PM to 9 PM) to improve overall efficiency, and investigate the factors driving the high response rate during these late hours.

OPTIMIZATION PLANNING & ACTIONABLE RECOMMENDATIONS



Optimization Planning & Actionable Recommendations

1. Strengthen Call Engagement Efficiency

- Audit dialing system, call scripts, and lead database to identify failure causes.
- Implement follow-up automation or multi-channel outreach (SMS/email) to recover unconnected leads.
- Apply call time segmentation to detect peak contact hours for specific regions.

2. Realign Outreach Timing

- Shift a portion of outreach attempts to evening hours (18:00–21:00), aligning with peak response patterns.
- Use dynamic scheduling and predictive models to continuously optimize call-hour distribution.

3. Balance Monthly Outreach Strategy

- Reduce excessive volume in low-performing months (June–July).
- Increase focus and resource allocation to high-response months (March–May) for better ROI.

4. Regional Targeting & Expansion

- Replicate successful engagement strategies from Canada and South Africa across similar markets.
- Conduct localized content testing for underperforming regions to improve global consistency.

5. Category-Level Optimization

- Strengthen Pre Admission outreach through personalized messaging and tailored communication strategies.
- Maintain effective Post Admission processes while introducing targeted nurturing for early-stage prospects.

Expected Outcomes



- +20–25% increase in overall response and engagement rates through better timing and channel optimization.
- Significant reduction in call failure rate via improved dialing accuracy and multi-channel follow-up.
- Balanced monthly outreach performance, leading to more consistent results year-round.
- Expanded global engagement footprint, reducing overdependence on a few high-performing regions.
- Higher Pre Admission conversion potential, strengthening the overall admission pipeline.
- Improved operational efficiency and ROI, driven by data-backed decision-making and targeted optimization.

By implementing this plan, we can transform outreach into a high-impact, data-driven engine for sustained enrollment growth and measurable success.

CONCLUSION



Thus, the outreach analysis reveals valuable insights into engagement effectiveness across time, regions, and admission stages. Strategic timing, regional targeting, and process efficiency are identified as key factors influencing overall response and conversion rates. These findings provide a strong foundation for data-driven decision-making, enabling optimized outreach strategies and improved future campaign performance.

We, the team, sincerely thank Excelerate, our department head, and associates for the opportunity and guidance during the Early Internship program, which greatly enhanced our analytical and visualization skills.

THANK YOU

excelerate



The Virtual Presentation Recording Link:

<https://drive.google.com/file/d/1UZCPRr7TNcSyhrob8VpWRyZCZO-sDiMZ/view?usp=sharing>